CentOS 7 Mail Server Setup

Installation, Configuration, and Testing Guide

This document provides a step-by-step guide to installing, configuring, and testing a mail server on CentOS 7. Postfix is used for mail transfer (SMTP), while Dovecot provides mailbox access services. The setup is intended for a controlled local-network lab and is tested using a client user.

1. Prerequisites

- CentOS 7 installed on a physical machine or virtual machine.
- Root access or a user with sudo privileges.
- Internet connectivity for package installation.
- Basic familiarity with Linux terminal commands.

2. Step I: Log in as Root

If logged in as a regular user (for example, sam), switch to the root account:

```
su -
```

Enter the root password when prompted.

3. Step II: Verify Internet Connectivity

Check whether the server can reach an external IP address:

```
ping -c 4 8.8.8.8
```

If IP connectivity works but DNS resolution fails, edit the resolver configuration:

```
vi /etc/resolv.conf
```

Add the following DNS server entries if appropriate for your network:

```
nameserver 8.8.8.8
nameserver 1.1.1.1
```

In vi, press Esc, type :wq, and press Enter to save and exit. Note that NetworkManager may overwrite /etc/resolv.conf; configure DNS persistently through the network connection if needed.

4. Step III: Configure CentOS 7 Vault Repositories

CentOS 7 is end-of-life, so its standard mirror infrastructure is no longer maintained. For a lab system, configure the archived CentOS 7.9.2009 Vault repositories. These archives do not provide ongoing security updates.

```
mkdir -p /etc/yum.repos.d/backup
mv /etc/yum.repos.d/CentOS-* /etc/yum.repos.d/backup/
```

Create the repository file:

```
cat > /etc/yum.repos.d/CentOS-Vault.repo <<'EOF'
[base]
name=CentOS-7-Base
baseurl=http://vault.centos.org/7.9.2009/os/$basearch/
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7
```

```
[updates]
name=CentOS-7-Updates
baseurl=http://vault.centos.org/7.9.2009/updates/$basearch/
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7

[extras]
name=CentOS-7-Extras
baseurl=http://vault.centos.org/7.9.2009/extras/$basearch/
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7
EOF
```

Refresh and verify repository metadata:

```
yum clean all
yum repolist all
```

5. Step IV: Install Postfix and Dovecot

```
yum install -y postfix dovecot
```

Verify the installed versions:

```
postconf -d | grep mail_version
dovecot --version
```

6. Step V: Configure Postfix

Open the main Postfix configuration file:

```
vi /etc/postfix/main.cf
```

Set or update the following directives. Replace example.com with the domain used in your lab:

```
myhostname = mail.example.com
mydomain = example.com
myorigin = $mydomain
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
inet_interfaces = all
inet_protocols = ipv4
home_mailbox = Maildir/
```

Ensure each directive appears only once as an active setting. Save and exit with Esc, then :wq, then Enter.

7. Step VI: Start and Enable Postfix

```
systemctl start postfix
systemctl enable postfix
systemctl status postfix
```

Confirm that the service reports Active: active (running).

8. Step VII: Configure Dovecot

Edit the mail location configuration:

```
vi /etc/dovecot/conf.d/10-mail.conf
mail_location = maildir:~/Maildir
```

Edit the authentication configuration:

```
vi /etc/dovecot/conf.d/10-auth.conf
disable_plaintext_auth = no
auth_mechanisms = plain login
```

These authentication settings are suitable only for a controlled lab with trusted network access. Plaintext authentication should not be exposed on an untrusted network without TLS. Save each file with Esc, :wq, and Enter.

9. Step VIII: Start and Enable Dovecot

```
systemctl start dovecot
systemctl enable dovecot
systemctl status dovecot
```

Confirm that the service reports Active: active (running).

10. Step IX: Configure Firewall and SELinux

Open the ports used in this lab. Port 25 is SMTP; ports 110 and 143 are POP3 and IMAP; port 993 is IMAPS. Port 110 is unencrypted POP3, and port 143 is typically unencrypted IMAP unless TLS is configured.

```
firewall-cmd --permanent --add-port=25/tcp
firewall-cmd --permanent --add-port=110/tcp
firewall-cmd --permanent --add-port=143/tcp
firewall-cmd --permanent --add-port=993/tcp
firewall-cmd --reload
```

If SELinux is enforcing, apply the setting only if required by the actual delivery configuration and after checking the relevant SELinux denials:

```
setsebool -P allow_postfix_local_write_mail_spool on
```

For production use, prefer a supported operating system, TLS certificates, secure authentication, and only the firewall ports that are genuinely required.

11. Step X: Test the Mail Server Locally

Send a local test message to root:

```
echo "Hello Mail Server" | mail -s "Test Subject" root@localhost
```

Monitor the mail log:

```
tail -f /var/log/maillog
```

A successful SMTP acceptance is a useful first check, but it does not by itself prove that remote delivery or client mailbox access works. Verify the log for delivery status and inspect the recipient mailbox.

12. Step XI: Create a Client User for Testing

```
useradd client1
```

```
passwd client1
```

Enter and confirm a password when prompted. Switch to the client account:

```
su - client1
```

Send a test message to the local root account:

```
echo "Hello from client" | mail -s "Client Test" root@localhost
```

If the mail command is not installed, install the package that provides it (commonly mailx on CentOS 7):

```
yum install -y mailx
```

13. Step XII: Monitor Live Mail Traffic

Open a separate terminal, switch to root if required, and monitor mail activity in real time:

```
tail -f /var/log/maillog
```

Press Ctrl+C to stop following the log.

14. Troubleshooting Checklist

- Check service status with systemctl status postfix and systemctl status dovecot.
- Review /var/log/maillog for SMTP, authentication, and delivery errors.
- Check listening ports with ss -lntp.
- Verify DNS, hostname, and /etc/hosts settings for the lab domain.
- Confirm firewall rules and SELinux denials if connections or delivery fail.
- Ensure Postfix and Dovecot use compatible mailbox locations and permissions.

15. Important Security Notes

CentOS 7 is end-of-life and should not be used for an internet-facing production mail server. This guide is intended for an isolated lab or educational environment. Do not expose plaintext authentication or unencrypted POP3/IMAP to untrusted networks. For production, migrate to a currently supported operating system and configure TLS, secure authentication, DNS records (MX, A/AAAA, PTR as applicable), anti-abuse controls, and monitoring.